

Top 100 Linux Debug Commands Explained

By Devops Shack



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SYSTEM INFORMATION & OS DETAILS

1. `uname -a` → Shows kernel name, version, architecture. *(Check kernel compatibility or bug reports)*
➤ `uname -a`
2. `hostnamectl` → Displays hostname, OS, and kernel version. *(Quick OS summary)*
➤ `hostnamectl`
3. `lsb_release -a` → Distro version and release info. *(Used to confirm repo compatibility)*
➤ `lsb_release -a`
4. `cat /etc/os-release` → OS info file. *(Universal across distros)*
➤ `cat /etc/os-release`
5. `uptime` → System uptime + load averages. *(Diagnose sustained system load)*
➤ `uptime`
6. `whoami` → Current logged-in user. *(For permission verification)*
➤ `whoami`
7. `id` → Displays UID, GID, and group membership. *(Helps with permission debugging)*
➤ `id`
8. `dmesg -T | less` → Kernel messages with timestamps. *(Driver errors, disk/NIC issues)*
➤ `dmesg -T | grep error`

9. `journalctl -k -b` → Kernel log since boot. *(Detect boot-level issues)*

➤ `journalctl -k -b`

10. `hostname -I` → System IPs. *(Useful for multi-NIC servers)*

➤ `hostname -I`

CPU, MEMORY & PROCESS DEBUG

11. `top` → Live CPU, RAM, and process monitoring.

➤ `top`

12. `htop` → Advanced version of `top` with colors & search.

➤ `htop`

13. `vmstat 1` → CPU/memory/swap overview every 1 sec. *(Spot bottlenecks)*

➤ `vmstat 1`

14. `sar -u 1 5` → Historical CPU stats. *(Find recurring spikes)*

➤ `sar -u 1 5`

15. `mpstat -P ALL 1` → Per-core CPU load. *(Identify single-thread overload)*

➤ `mpstat -P ALL 1`

16. `free -h` → Memory & swap usage. *(Detect memory leaks)*

➤ `free -h`

17. `ps aux --sort=-%cpu | head` → Top CPU consumers.

➤ `ps aux --sort=-%cpu | head`

18. `pidstat -p <pid> 1` → Per-process CPU, memory, I/O usage.

➤ `pidstat -p 1234 1`

19. `pstree -pa` → Process hierarchy. *(Trace parent-child relations)*

➤ `pstree -pa`

20. `nice / renice` → Change process CPU priority. *(Fix starving processes)*

➤ `renice -n 10 -p 1234`

21. `kill -9 <pid>` → Force kill hung processes.

➤ `kill -9 4567`

22. `ulimit -a` → Display user resource limits. *(Useful for ulimit errors)*

➤ `ulimit -a`

23. `oomctl / journalctl | grep -i oom` → Check for OOM kills.

➤ `journalctl -k | grep -i oom`

24. `ps -eo pid,cmd,etime` → Process elapsed time. *(Long-running process detection)*

➤ `ps -eo pid,cmd,etime`

25. `lsof -p <pid>` → List open files by a process. *(Memory/file descriptor leaks)*

➤ `lsof -p 1234`

DISK & FILESYSTEM DEBUG

26. `df -hT` → Filesystem type and usage. *(Full disk detection)*

➤ `df -hT`

27. `du -sh / | sort -h*` → Top space consumers.

➤ `du -sh /var/* | sort -h`

28. `lsblk -f` → View block devices, labels, FS types.

➤ `lsblk -f`

29. `blkid` → Show UUIDs and labels of drives. *(fstab troubleshooting)*

➤ `blkid`

30. `mount | column -t` → Display mounted filesystems.

➤ `mount | column -t`

31. `findmnt /path` → Check where a directory is mounted.

➤ `findmnt /var`

32. `iostat -xz 1` → Disk I/O performance (reads/writes, queue, latency).

➤ `iostat -xz 1`

33. **iostat -oPa** → Monitor live per-process disk usage.

➤ **iostat -oPa**

34. **smartctl -a /dev/sda** → Disk health check. (*Detect failing drives*)

➤ **smartctl -a /dev/sda**

35. **hdparm -Tt /dev/sda** → Test read/write speed.

➤ **hdparm -Tt /dev/sda**

36. **fsck -n /dev/sda1** → Check filesystem consistency (non-destructive).

➤ **fsck -n /dev/sda1**

37. **xfs_repair -n /dev/sda1** → Check XFS file systems.

➤ **xfs_repair -n /dev/sda1**

38. **btrfs scrub status /** → Check Btrfs integrity.

➤ **btrfs scrub status /**

39. **lsof +D /path** → See which files are open in a directory.

➤ **lsof +D /var/log**

40. **fuser -vm /mnt/data** → Identify which process is using a mount.

➤ **fuser -vm /mnt/data**

NETWORK DEBUG

41. **ip addr** → View network interfaces.

➤ **ip addr**

42. **ip route** → Routing table & gateway check.

➤ **ip route**

43. **ss -tulpn** → Listening ports and processes. (*Modern netstat*)

➤ **ss -tulpn**

44. **ss -s** → Socket summary (TIME_WAIT, ESTAB).

➤ **ss -s**

45. **ping -c 4 host** → Check network reachability.

➤ `ping -c 4 8.8.8.8`

46. `tracert host` → Trace route hops. (*Network delay localization*)

➤ `tracert google.com`

47. `mtr -rw host` → Live route & packet loss test.

➤ `mtr -rw 1.1.1.1`

48. `dig +short example.com` → DNS resolution test.

➤ `dig +short example.com`

49. `nslookup example.com` → Alternative DNS test.

➤ `nslookup example.com`

50. `curl -v URL` → Verbose HTTP test. (*TLS, redirects, proxy*)

➤ `curl -v https://devopsshack.com`

51. `wget --server-response URL` → Inspect HTTP headers.

➤ `wget --server-response https://site.com`

52. `nc -vz host port` → Check TCP connectivity. (*Firewall debugging*)

➤ `nc -vz 10.0.0.1 22`

53. `telnet host port` → Test open ports (legacy).

➤ `telnet 10.0.0.1 443`

54. `ethtool eth0` → Check NIC link speed & errors.

➤ `ethtool eth0`

55. `ifconfig -a` → Legacy network info. (*Still used on older servers*)

➤ `ifconfig -a`

56. `nmcli device status` → NetworkManager device summary.

➤ `nmcli device status`

57. `ip neigh` → Show ARP table. (*Neighbor resolution*)

➤ `ip neigh`

58. `nmap -sT -p 1-1024 host` → Scan open ports. (*Security debugging*)

➤ `nmap -sT -p 22,80,443 localhost`

59. `tcpdump -i eth0 port 80` → Capture packets on port.

➤ `tcpdump -nn -i eth0 port 443`

60. `ngrep -d any 'Host:' tcp port 80` → Filtered packet view.

➤ `ngrep -d any 'Host:' tcp port 80`

61. `iperf3 -c host` → Bandwidth test.

➤ `iperf3 -c server_ip`

62. `netstat -rn` → Routing table summary. (*Legacy check*)

➤ `netstat -rn`

63. `ss -tan state established` → Show active TCP connections.

➤ `ss -tan state established`

64. `arp -n` → ARP table cache. (*MAC resolution issues*)

➤ `arp -n`

65. `route -n` → Old-school routing check.

➤ `route -n`

LOGS, SERVICES & BOOT

66. `journalctl -xe` → View system logs with errors.

➤ `journalctl -xe`

67. `journalctl -u nginx` → Service-specific logs.

➤ `journalctl -u nginx`

68. `systemctl status nginx` → Service health & last log lines.

➤ `systemctl status nginx`

69. `systemctl list-units --type=service` → All active services.

➤ `systemctl list-units --type=service`

70. `systemctl cat nginx` → View exact unit file content.

➤ `systemctl cat nginx`

71. `systemd-analyze blame` → Show boot time breakdown.

➤ `systemd-analyze blame`

72. `systemd-analyze critical-chain` → Dependency boot chain.

➤ `systemd-analyze critical-chain`

73. `loginctl list-sessions` → Active user sessions.

➤ `loginctl list-sessions`

74. `last -10` → Last 10 user logins.

➤ `last -10`

75. `who / w` → Logged-in users and activity.

➤ `who`

76. `dmesg | tail -20` → Latest kernel events.

➤ `dmesg | tail -20`

77. `lsmod` → Loaded kernel modules. (*Driver troubleshooting*)

➤ `lsmod`

78. `modinfo <module>` → Module version & parameters.

➤ `modinfo e1000`

79. `insmod / rmmod` → Load/unload kernel modules. (*Advanced debugging*)

➤ `rmmod <module>`

80. `journalctl -b -1` → Previous boot logs.

➤ `journalctl -b -1`

FILES, TEXT & PERMISSIONS

81. `find /path -type f -mtime -1` → Files changed in last day.

➤ `find /var/log -type f -mtime -1`

82. `grep -Rin "error" /var/log` → Recursively search logs.

➤ `grep -Rin "Connection refused" /var/log`

83. `awk '{print $1,$7}' access.log | sort | uniq -c | sort -nr` → Parse logs.

➤ Log summarization.

84. `diff file1 file2` → Compare configs.

➤ `diff nginx.conf nginx.conf.bak`

85. `stat file` → Show file timestamps, inode.

➤ `stat /etc/passwd`

86. `file file.bin` → Detect file type. (*ELF, text, compressed*)

➤ `file ./binary`

87. `md5sum / sha256sum file` → Verify integrity.

➤ `sha256sum download.iso`

88. `tar -tvf archive.tar.gz` → List archive contents.

➤ `tar -tvf backup.tar.gz`

89. `gzip -d file.gz / zcat file.gz` → Decompress logs.

➤ `zcat syslog.1.gz`

90. `chattr +i / lsattr file` → File immutability check.

➤ `lsattr /etc/hosts`

91. `chmod / chown` → Fix permissions/ownership.

➤ `chmod 644 /var/log/app.log`

92. `lsof +L1` → Detect deleted files still open. (*Disk full mystery fix*)

➤ `lsof +L1`

93. `wc -l file` → Count lines. (*Log size check*)

➤ `wc -l error.log`

94. `head / tail -n` → Quick file preview.

➤ `tail -n 20 /var/log/messages`

95. `tee file.log` → Display and save output.

➤ `ping 8.8.8.8 | tee ping.log`

ADVANCED DEBUGGING & PERFORMANCE

96. `strace -f -p <pid>` → Trace syscalls. (*Hang/Crash root cause*)

➤ `strace -f -p 1234`

97. `ltrace ./binary` → Trace library calls. (*Runtime issues*)

➤ `ltrace ./app`

98. `perf top` / `perf stat` → Performance profiling.

➤ `perf stat ./program`

99. `time cmd` → Measure command execution time.

➤ `time ./deploy.sh`

100. `sar -n DEV 1 3` → Network throughput stats. (*Traffic analysis*)

➤ `sar -n DEV 1 3`